AI & DATA SCIENCE TRACK

Healthcare Predictive Analytics

AI-Powered Diagnosis Assistant System

Disease Diagnosis | Cancer Gene Classification

Project Type	AI & Machine Learning — Healthcare Domain
Systems	(1) Disease Diagnosis from Symptoms | (2) Cancer Type from Gene Expression
Track	AI & Data Science
Document Version	v1.0

1. Project Planning & Management

1.1 Project Proposal

Overview

The Healthcare Predictive Analytics system is a dual-module AI platform designed to assist medical professionals in clinical decision-making. The platform integrates two complementary machine learning pipelines:

Module A — Disease Diagnosis Assistant
Accepts patient-reported symptoms as input and returns the most probable disease(s) using a multi-class classifier trained on 246,945 patient records across 773 diseases and 377 symptom features.

Module B — Cancer Gene Expression Classifier
Accepts RNA-seq gene expression profiles from The Cancer Genome Atlas (TCGA) Pan-Cancer dataset and classifies the cancer type (33 types) using dimensionality-reduced features and supervised ML models.

Objectives

- Develop an accurate, real-time disease prediction system from symptom inputs.
- Build a robust cancer-type classifier using high-dimensional gene expression data.
- Design a unified, doctor-friendly interface for both prediction modules.
- Achieve high model performance (target F1-score ≥ 0.85 on held-out test sets).
- Ensure system scalability, maintainability, and clinical interpretability.

Project Scope

In Scope	Out of Scope
EDA, preprocessing, and model training	EMR / hospital system integration
Disease prediction from symptoms (773 diseases)	Real-time patient monitoring
Cancer classification from gene expression (33 types)	Drug recommendation engine
A functional standalone application for real-time diagnostic simulation	Full-scale production deployment on cloud servers
Model evaluation and interpretability analysis	Mobile native applications

1.2 Project Plan — Timeline & Milestones

Gantt Chart Summary

Milestone / Deliverable	Week 1	Week 2	Week 3	Week 4	Owner
M1: Data Collection & Exploration	■				Data Engineer
M2: Preprocessing & EDA Visualizations	■	■			Data Scientist
M3: Model Training & Evaluation		■	■		ML Engineer
M4: System Integration & UI			■	■	Full-Stack Dev
M5: Testing, Documentation & Presentation				■	All Team

Legend: ■ = Active development phase | Columns represent project weeks

Detailed Milestones & Deliverables

#	Milestone	Deliverables	Duration	Status
1	Data Collection & Exploration	EDA Notebook, Dataset Exploration Report, Cleaned CSV	Week 1	Completed
2	Data Analysis & Visualization	Analysis Report, Interactive Plotly Dashboards	Weeks 1-2	Completed
3	Model Development	Trained Models (RF, XGBoost), Evaluation Report, Confusion Matrices	Weeks 2-3	In Progress
4	System Integration & UI	Web App Prototype, API Endpoints, User Interface	Weeks 3-4	Planned
5	Testing & Documentation	Test Report, Final Documentation, Presentation Slides	Week 4	Planned

Resource Allocation

Resource	Type	Purpose
Python (pandas, sklearn, plotly)	Software	Data processing, ML model training, visualization
Jupyter Notebook / Google Colab	Platform	Interactive development and experimentation
TCGA Pan-Cancer Gene Dataset	Data	Cancer classification training (~10K samples, 20K+ genes)
Disease-Symptoms Dataset	Data	Disease diagnosis training (246,945 records, 773 diseases)
Simple Desktop/Web GUI	Software	Provides a lightweight and direct interface for symptom selection and gene data processing
GitHub	Tool	Version control and team collaboration

1.3 Task Assignment & Team Roles

Role	Responsibilities	Key Tasks	Milestone Owner
Project Manager	Planning, coordination, documentation	Gantt chart, risk log, stakeholder updates, final report	M1, M5
Data Engineer	Data acquisition, cleaning, pipeline	Load TCGA + symptoms datasets, handle missing values, feature engineering	M1, M2

Role	Responsibilities	Key Tasks	Milestone Owner
Data Scientist	EDA, visualization, statistical analysis	EDA notebooks, heatmaps, PCA plots, correlation analysis, Plotly dashboards	M1, M2
ML Engineer	Model architecture, training, tuning	RF / XGBoost / Neural Network training, hyperparameter tuning, evaluation metrics	M3
Full-Stack Developer	UI/UX design, API, deployment	Doctor-facing web interface, symptom input form, gene upload, prediction display	M4
QA / Tester	Testing and quality assurance	Unit tests, integration tests, edge-case validation, UAT with medical scenarios	M5

1.4 Risk Assessment & Mitigation Plan

Risk	Probability	Impact	Mitigation Strategy	Owner
Class imbalance in disease dataset	High	High	Apply SMOTE oversampling; use weighted F1-score evaluation	ML Engineer
Local system memory constraints when processing 20K+ gene features	High	High	Optimized data loading using chunking and efficient PCA transformation to ensure smooth performance on standard hardware	ML Engineer
Model overfitting on training data	Medium	High	Cross-validation (5-fold), regularization, early stopping	Data Scientist
Slow inference time with large models	Medium	Medium	Model compression, caching predictions, async API calls	Developer
Insufficient team availability	Low	Medium	Define backup owners per task; weekly sync meetings	PM
Data privacy / HIPAA compliance concerns	Low	High	Use only de-identified, publicly available datasets (TCGA, Kaggle)	PM

1.5 Key Performance Indicators (KPIs)

Model Performance KPIs

KPI Metric	Module A (Symptoms)	Module B (Genes)	Measurement Method
Accuracy	≥ 88%	≥ 85%	Test set evaluation (sklearn metrics)
Weighted F1-Score	≥ 0.85	≥ 0.82	Per-class F1 weighted by support
Top-3 Accuracy	≥ 95%	≥ 92%	Disease in top 3 predictions
Precision (Macro)	≥ 0.80	≥ 0.80	Macro-averaged precision
Recall (Macro)	≥ 0.80	≥ 0.78	Macro-averaged recall

System Performance KPIs

KPI	Target	Description
Response Time (Symptom Query)	< 2 seconds	Time from symptom input to prediction display
Response Time (Gene Upload)	< 10 seconds	End-to-end gene preprocessing + classification time
System Uptime	≥ 99%	Availability during demo and evaluation period
User Adoption Rate (Demo)	≥ 80% task completion	% of evaluators who successfully complete a diagnosis task
Error Rate	< 1%	System/API errors per 100 requests during testing

2. Literature Review

2.1 Related Work & Background

Disease Diagnosis from Symptom Data

Machine learning-based clinical decision support systems (CDSS) have demonstrated significant effectiveness in automating disease diagnosis. Studies using symptom-based classifiers on structured datasets have achieved accuracies exceeding 90% for common disease categories. Random Forest and Gradient Boosting models consistently outperform traditional rule-based expert systems due to their ability to capture non-linear symptom interactions.

Cancer Classification from Gene Expression

The TCGA Pan-Cancer Atlas project established the gold standard for multi-cancer genomic classification. Deep learning approaches applied to gene expression data have achieved state-of-the-art performance exceeding 95% accuracy on cancer-type classification tasks. Dimensionality reduction via PCA or autoencoders is essential given the typical 20,000+ feature space of RNA-seq data.

2.2 Feedback & Evaluation Criteria

Evaluation Framework

Assessment Category	Weight	Criteria
Documentation & Planning	20%	Project proposal clarity, risk assessment completeness, requirements detail
Data Collection & EDA	20%	Dataset quality, EDA depth, visualization quality, insight derivation
Model Development & Performance	30%	Model selection rationale, achieved metrics, comparison of approaches
System Implementation	15%	Working prototype, UI/UX quality, integration of both modules
Testing & Validation	10%	Test coverage, edge cases, error handling, validation on realistic scenarios
Final Presentation	5%	Clarity, depth, ability to answer questions, demo quality

Suggested Improvements (Pre-Submission Checklist)

- Add confidence score / probability distribution alongside each prediction for clinical transparency.
- Include SHAP or LIME explainability outputs to show which symptoms/genes drove each prediction.
- Extend Module A to handle partial symptom sets gracefully (not all 377 symptoms will be available).
- For Module B, benchmark at least 3 model architectures (RF, XGBoost, MLP) and report comparative results.
- Add a confusion matrix heatmap visualization per module in the final presentation.
- Document API endpoints with request/response schemas for reproducibility.

3. Requirements Gathering

3.1 Stakeholder Analysis

Stakeholder	Type	Needs & Expectations	Engagement Level
General Practitioners (GPs)	Primary User	Fast symptom-to-diagnosis lookup; confidence scores; top-3 results	High — Direct User
Oncologists	Primary User	Accurate cancer-type classification from gene profiles; biomarker highlights	High — Direct User
Hospital IT Administrators	Technical	System security, HIPAA compliance, integration APIs, uptime guarantees	Medium — Integration
Lab Technicians / Bioinformaticians	Secondary User	Gene expression data upload interface; batch processing capability	Medium — Power Users
Project Lecturers / Evaluators	Academic	Code quality, documentation completeness, model performance metrics	High — Assessors
AI Development Team	Internal	Clear requirements, clean data pipelines, reproducible experiments	High — Builders

3.2 User Stories & Use Cases

Module A — Disease Diagnosis (Symptom-Based)

ID	As a...	I want to...	So that...
US-01	Doctor	Select multiple symptoms from a searchable list	I can quickly enter patient-reported symptoms without free typing
US-02	Doctor	View the top 3 predicted diseases with probability scores	I can prioritize which diagnosis to investigate further
US-03	Doctor	See which symptoms were most influential in each prediction	I understand the reasoning behind the AI recommendation
US-04	Doctor	Clear all inputs and start a new diagnostic session	Each patient session is independent and error-free

Module B — Cancer Classification (Gene Expression)

ID	As a...	I want to...	So that...
US-05	Oncologist	Upload a gene expression file (CSV/TSV) for a patient sample	I do not have to manually input thousands of gene values
US-06	Oncologist	Receive the predicted cancer type with confidence score	I have AI-aided support in confirming the cancer classification
US-07	Bioinformatician	See the top expressed biomarker genes that drove the prediction	I can validate the result against known cancer signatures
US-08	IT Admin	Monitor system logs and error reports via admin dashboard	I can ensure system reliability and audit trail compliance

Use Case Diagram Summary

Primary Use Cases:
UC-1: Doctor enters symptoms → System returns ranked disease predictions
UC-2: Doctor uploads gene expression file → System classifies cancer type
UC-3: Doctor views prediction explanation (feature importance / SHAP)
UC-4: Admin accesses system logs and performance dashboard

Actors: Doctor (GP / Specialist), Bioinformatician, IT Administrator, AI System

3.3 Functional Requirements

FR-A: Disease Diagnosis Module

ID	Requirement	Description	Priority
FR-A1	Symptom Input Interface	System shall provide a searchable multi-select interface for all 377 available symptoms	Must Have
FR-A2	Disease Prediction Output	System shall return the top-3 most likely diseases with associated probability scores for any symptom set >= 1 symptom	Must Have
FR-A3	Feature Importance Display	System shall highlight the top-5 symptoms contributing most to each predicted disease	Should Have
FR-A4	Session Reset	System shall allow users to clear all selected symptoms and reset predictions with a single action	Must Have
FR-A5	Model Backend	System shall use a pre-trained Random Forest or XGBoost model loaded at startup with < 2s inference latency	Must Have

FR-B: Cancer Gene Classification Module

ID	Requirement	Description	Priority
FR-B1	Gene File Upload	System shall accept CSV/TSV gene expression files and validate format before processing	Must Have
FR-B2	Automatic Preprocessing	System shall automatically apply PCA transformation (matching training pipeline) before inference	Must Have
FR-B3	Cancer Type Prediction	System shall return predicted cancer type (from 33 TCGA types) with full name and confidence score	Must Have
FR-B4	Biomarker Highlighting	System shall display top-10 most influential genes for the predicted cancer type	Should Have
FR-B5	Label Mapping	System shall display both TCGA abbreviation (e.g., BRCA) and full clinical name (Breast Invasive Carcinoma)	Should Have

3.4 Non-Functional Requirements

Performance

ID	Requirement	Specification	Priority
NFR-P1	Symptom Prediction Latency	End-to-end response time shall be < 2 seconds for any symptom input set	Critical
NFR-P2	Gene Classification Latency	Full pipeline (upload + preprocess + classify) shall complete within 10 seconds	High
NFR-P3	Concurrent Users	System shall support at least 10 concurrent users without performance degradation	Medium

Security

ID	Requirement	Specification	Priority
NFR-S1	Data Anonymization	No personally identifiable patient information (PII) shall be stored or transmitted	Critical
NFR-S2	Input Validation	All inputs shall be validated and sanitized before processing to prevent injection attacks	High
NFR-S3	Session Management	User sessions shall auto-expire after 30 minutes of inactivity	Medium

Usability

ID	Requirement	Specification	Priority
NFR-U1	Learnability	A new medical user shall complete a diagnostic task within 5 minutes without training	High
NFR-U2	Accessibility	Interface shall be accessible via modern browsers (Chrome, Firefox, Edge) without plugin installation	High
NFR-U3	Error Feedback	System shall display clear, actionable error messages for invalid inputs (e.g., wrong file format)	High
NFR-U4	Language Support	Cancer type labels shall be displayed in both English and Arabic for broader clinical accessibility	Medium

Reliability

ID	Requirement	Specification	Priority
NFR-R1	System Uptime	System shall maintain >= 99% uptime during the evaluation and testing period	High
NFR-R2	Graceful Degradation	System shall display a meaningful error message (not a crash) for any unsupported input	High
NFR-R3	Reproducibility	Identical inputs shall always produce identical prediction outputs (deterministic inference)	Critical

Priority Legend:
Critical = Must be met for project approval | High = Expected in final submission | Medium = Nice-to-have for bonus credit